

MICHAEL C. RUGGIERO

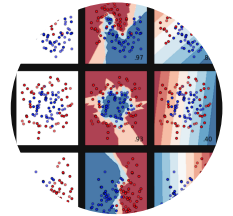
Data Scientist / Machine Learning Engineer

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EXPERIENCE

A/AI Machine Learning Engineer, Staff

Lockheed Martin

August, 2021 – Present

Remote

- **Lead Data Scientist:** Responsible for ACE team AI implementations.
- **RAG/AI Design:** Designed AI agent for CI hub.
- **ELK, Python:** Built Transformer enabled DIS simulation with ELK stack.
- **Mentoring:** Manages capstone programs and team lit review.
- **Oversight:** Interviewed programmers, managing the code of developers.
- **Dash:** Built NLP app with spaCy to analyze Powerpoints slides.

Graphical Machine Learning Engineer

iRobot

April, 2020 – May, 2021

Remote

- **3D Modeling:** Built physics and lighting integration into room simulation.
- **CUDA, Python:** Designed automated fluid simulation with textures.
- **Slack, AWS:** Managed projects with multiple team members remotely.
- **CUDA, Pytorch:** Built CycleGAN model to generate visual artifacts.
- **Presentation:** Enabled effective decision making with the organization.

Data Scientist

Campaign to Elect Breanna Lungo-Koehn

March, 2019 – February, 2020

Medford, Massachusetts

- **Scikit-learn:** Charted and mapped canvassing strategies.
- **Python, SQL:** Built interactive database for non-technical volunteers.
- **Geolocating:** Built interactive ArcGIS visualizations to chart progress.
- **Management:** Organized routing and phone banking for canvassers.
- **NLP:** Scrapped partisan facebook/subreddits for essential information.

Data Science Immersive

General Assembly

February, 2019 – June, 2019

Boston, Massachusetts

- **Networkx:** Built graph representation for all Medford roads and traffic.
- **Pytorch:** Designed reinforcement learning AI to play pong.

Machine Learning Engineer

Lay's Potato Chips

March, 2015 – March, 2017

Remote Contract

- **Python, C#:** Incorporated tensors, inertia, and friction into the engine.
- **Data Management:** Scanned chips and programmed mesh systems.

SKILLS

Programming

Python

C# / C++

Deep NLP

Hugging Face

Fine-tuning

Clustering

KNN

Spectral

NLP

Langchain

RAG Vectorization

Machine Learning

Pytorch

Sci-Kit learn

Data Processing

SQL

Elastic Search

Visualization

Streamlit

Dash

Cloud

AWS

Docker

EDUCATION

Advanced Studies Fellow

Massachusetts Institute of Technology

Sep 2013 – May 2014

Masters: English

Temple University

Sept 2010 – May 2012

B.Sc. Mathematics

Rhode Island College

Sept 1999 – May 2003

ACHIEVEMENTS



Elected to School Committee

Used k-mean clustering to organize 2017 campaign canvassing.



Automated Farming

Using Arduino sensors, built 2 acre organic, blueberry farm.